Campus Food Ordering Platform

Project Plan

Version 1.0 | Agile Delivery | 4-Week Sprint Cycle

1. Project Overview

The Campus Food Ordering Platform is a web-based application designed to eliminate long queue times at university food courts. Students can browse multi-vendor menus, place and pay for orders online, and collect meals with minimal wait time. Vendors manage their menus and order queues through a dedicated portal, while administrators oversee the platform through an approval and analytics dashboard.

1.1 Objectives

- Implement Agile methodology with iterative 1-week sprints over a 4-week delivery cycle.
- Apply CI/CD principles using GitHub Actions and Firebase Hosting for automated deployments.
- Follow a test-driven development (TDD) approach throughout all feature development.
- Deliver a publicly accessible, production-ready web application.

1.2 Scope

- Multi-vendor menu browsing, ordering, and payment via Paystack.
- Real-time order status tracking with email notifications.
- Order cancellation (before preparation begins) and Paystack refund processing.
- Vendor menu management including allergen and dietary information standardised against the Open Food Facts reference dataset.
- Admin portal for vendor approval, suspension, and platform analytics.
- Analytics dashboards exportable as CSV or PDF.

2. Technology Stack

Technologies were selected to meet real-time data, scalability, and rapid delivery requirements:

Layer	Technology	Justification
Frontend	HTML, CSS, JavaScript	Lightweight; no build tooling overhead; fast iterations
Database	Google Firestore	Real-time listeners; scalable; native Firebase Auth integration
Authentication	Firebase Auth	Multi-role support; secure token-based auth; minimal setup
File Storage	Firebase Storage	Integrated with Firestore; hosts vendor menu photos
Backend Logic	Firebase Cloud Functions	Serverless; Firestore-triggered; handles Paystack webhooks & email
Payments	Paystack	South Africa-supported; card & EFT; webhook-based confirmation
Notifications	Email via Cloud Functions	Order-ready and cancellation/refund alerts sent to students
Hosting	Firebase Hosting	CDN-backed; HTTPS; CI/CD auto-deploy on merge
CI/CD	GitHub Actions	Automated test runs + Firebase deploy on merge to main
Dietary Data	Open Food Facts (reference)	Internationally recognised standard; used to define and normalise allergen & dietary categories across all vendors

3. Project Milestones & Delivery Plan

The project followed a 4-week Agile sprint cycle. Each sprint delivered a tested, deployable slice of functionality.

ID	Milestone	Timeline	Deliverables	Status
M1	Sprint 1 – Project Setup & Auth	Week 1	Firebase project, Auth roles (Student/Vendor/Admin), CI/CD pipeline	Completed
M2	Sprint 2 – Menu Management	Week 2	Vendor menu CRUD, allergen/dietary tags	Almost Completed
M3	Sprint 3 – Order Management & Payments	Week 3	Student ordering flow, Paystack integration, vendor order queue	Completed
M4	Sprint 4 – Tracking, Notifications & Cancellations	Week 3–4	Real-time status, email notifications, order cancellation	Completed
M5	Sprint 5 – Admin Portal & Analytics	Week 4	Vendor approval/suspension, 3 dashboard reports, CSV/PDF export	Completed
M6	Testing, QA & Deployment	Week 4	Full regression, TDD sign-off, Firebase Hosting deployment	Almost Completed

3.1 Agile Ceremonies

- Sprint Planning: Start of each sprint — define sprint backlog from product backlog.
- Daily Standups: Async team updates (blockers, progress, next steps).
- Sprint Review: End-of-sprint demo against acceptance criteria.
- Sprint Retrospective: Identify process improvements for the next sprint.

4. Team Roles & Responsibilities

- Product Owner: Maintained product backlog; prioritised features; accepted deliverables.
- Scrum Master: Facilitated ceremonies; removed blockers; tracked sprint velocity.
- Frontend Developers: HTML/CSS/JS; Firestore real-time integration; Paystack checkout UI.
- Backend Developers: Cloud Functions; Paystack webhook handler; email notification triggers.
- QA / Test Engineer: Unit and integration tests; regression testing; TDD sign-off.
- DevOps: GitHub Actions CI/CD pipeline; Firebase Hosting configuration and deployment.

5. Risk Register

Risk	Likelihood	Impact	Mitigation
Scope Creep	High	Medium	Strict sprint planning; backlog grooming each sprint
Firebase quotas exceeded under load	Medium	Low	Monitor reads/writes; implement query pagination
Paystack integration delays	Medium	Medium	Use sandbox environment early; buffer time in Sprint 3
Email notification deliverability	Low	Low	Verified sender domain; tested with multiple providers
Order cancellation & refund disputes	Medium	Medium	Define cancellation window policy; use Paystack Refund API

6. Quality Assurance & CI/CD

6.1 Test-Driven Development

- Unit tests written before feature implementation for all business logic.
- Integration tests covering Paystack payment flow, Firestore writes, and Cloud Function triggers.
- End-to-end tests for critical journeys: place order, cancel order, vendor updates status.

6.2 CI/CD Pipeline

- All changes submitted via Pull Requests to the main branch.
- GitHub Actions runs the full test suite on every PR — failed tests block merges.

- On merge to main: automated Firebase Hosting deployment to the live environment.

7. Key Design Decisions

7.1 Order Cancellation & Refund Policy

Students may cancel an order only while its status is 'Order Received' (before the vendor begins preparation). Once the vendor changes status to 'Preparing', cancellation is locked. On successful cancellation:

- The system calls the Paystack Refund API to initiate a full refund to the student's original payment method.
- The student receives an email confirmation of the cancellation and expected refund timeline.
- The vendor receives an email notification that the order was cancelled.
- The Firestore order document is updated to status 'Cancelled' with a timestamp and reason.

7.2 Dietary & Allergen Data Source

Open Food Facts was selected as the reference standard for allergen and dietary categorisation. Rather than calling an external API, the team used Open Food Facts as a recognised framework to define and normalise the allergen and dietary labels available to vendors when creating menu items. Justification:

- Internationally recognised standard, ensuring allergen and dietary categories are consistent and credible.
- Structured fields for allergens (nuts, gluten, dairy) and dietary labels (halal, vegan, vegetarian).
- Community-maintained with broad coverage of dietary and allergen classifications relevant to a South African university context.
- Used as a standardisation reference, not a live API integration — vendors select from a pre-defined, normalised list of allergen and dietary tags when creating menu items.

7.3 Real-time Order Tracking

Firestore real-time listeners on the student's tracking page instantly reflect vendor-driven status changes (Order Received → Preparing → Ready for Pickup) without page refresh. When status reaches 'Ready for Pickup', a Cloud Function dispatches an email notification to the student.

8. Deployment

- Application hosted on Firebase Hosting with a public URL.
- Firestore security rules enforce role-based access (student, vendor, admin).
- Paystack API keys and email credentials stored as Firebase environment config — never in source control.

- Production deployment triggered automatically on merge to main via GitHub Actions.

— End of Project Plan —